Problem 21 6.2.2025

a) 50 wavelength channels, 10 fiber modes to transmit a total data rate / bit rate of 200 $\frac{\text{Tbit}}{\text{s}}$

$\Rightarrow$ data rate per channel and mode: $R_b = \frac{200\,\frac{\text{Tbit}}{\text{s}}}{50 \cdot 10} = 400\,\frac{\text{Gbit}}{\text{s}}$

b) $OSNR|_{dB} = 58\,\text{dB} + P_0|_{dBm} - F_{amp}|_{dB} - \alpha_{dB} \cdot L_{oa} - 10 \cdot \log \frac{L}{L_{oa}} =$

$= 58\,\text{dB} + 10 \cdot \log \frac{10\,\text{mW}}{1\,\text{mW}} - 10 \cdot \log 4 - 0.2\,\frac{\text{dB}}{\text{km}} \cdot 100\,\text{km} - 10 \cdot \log \frac{6600\,\text{km}}{100\,\text{km}} =$

$= 23.78\,\text{dB}$

c) $R_s = R_b \cdot (1+OH) \cdot \frac{1}{2} \frac{1}{\log_2 M}$ ... symbol rate for M-QAM

$R_s \hat{\approx} \Delta f$ $\Rightarrow$ the higher M, the lower the bandwidth usage

From diagram: 16-QAM

$R_s = 400\,\frac{\text{Gbit}}{\text{s}} \cdot (1+0.07)\,\frac{1}{2}\,\frac{1}{\log_2 16} = 53.5\,\text{GBd}$

d) $SNR_{req,dB} = 16.5\,\text{dB}$ (from diagram)

$OSNR = \frac{\Delta f}{12.5\,\text{GHz}} \cdot SNR \;\hat{=}\; OSNR_{dB} = 10 \cdot \log \frac{\Delta f}{12.5\,\text{GHz}} + SNR_{dB}$

$R_s \hat{\approx} \Delta f = 53.5\,\text{GHz}$

$\Rightarrow OSNR_{req,dB} = 10 \cdot \log \frac{53.5\,\text{GHz}}{12.5\,\text{GHz}} + 16.5\,\text{dB} = 22.81\,\text{dB} \leq 23.78\,\text{dB}$

$\Rightarrow$ transmission link working

$\Delta f_{max} = 12.5\,\text{GHz} \cdot 10^{(OSNR_{dB} - SNR_{req,dB})/10} = 12.5\,\text{GHz} \cdot 10^{(23.78\,\text{dB} - 16.5\,\text{dB})/10}$

$= 64.13\,\text{GHz} \;\hat{\approx}\; R_s = 64.13\,\text{GBd}$

e) $OSNR_{dB} = 58\,dB + 10\cdot\log\frac{10mW}{1mW} - 10\cdot\log 4 - 0.2\frac{dB}{km}\cdot 120km - 10\cdot\log\frac{6600km}{120km} =$

$= 20.6\,dB < OSNR_{req,dB} = 22.81\,dB$

$\Rightarrow$ transmission link not working with 16-QAM

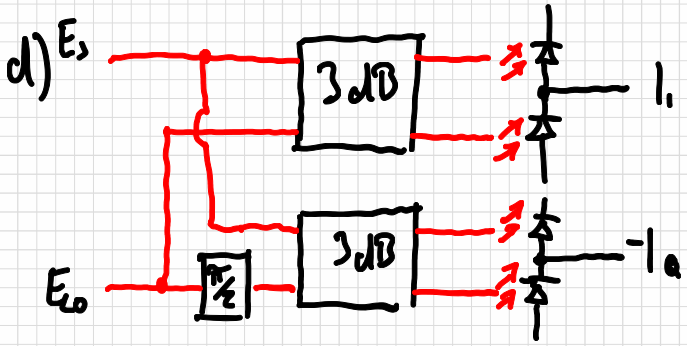
From diagram: 4-QAM at $SNR_{req,dB} = 9.8\,dB$

$R_s = 400\frac{Gbit}{s}(1+0.07)\frac{1}{2}\frac{1}{\log_2 4} = 107\,GBd \;\hat{\approx}\; \Delta f = 107\,GHz$

$OSNR_{req,dB} = 10\cdot\log\frac{107GHz}{12.5GHz} + 9.8\,dB = 19.12\,dB < OSNR_{dB} = 20.6\,dB$

$\Rightarrow$ transmission link is working by extending the bandwidth usage

d)



$E_1 = \frac{1}{2}(E_s + E_{LO})$

$E_2 = \frac{1}{2}(E_s - E_{LO})$

$E_3 = \frac{1}{2}(E_s + jE_{LO})$

$E_4 = \frac{1}{2}(E_s - jE_{LO})$

with $E_s = A_s(t)\,e^{j(\omega_s t + \theta_s(t))}$, $E_{LO} = A_{LO}\,e^{j\omega_{LO}t}$

$E_{1,2} = \frac{1}{2}\left(A(t)\,e^{j(\omega_s t + \theta_s(t))} \pm A_{LO}\,e^{j\omega_{LO}t}\right)$

$E_{3,4} = \frac{1}{2}\left(A(t)\,e^{j(\omega_s t + \theta_s(t))} \pm A_{LO}\,e^{j(\omega_{LO}t + \frac{\pi}{2})}\right)$

$I_1 = \mathcal{R}|E_1|^2 = \frac{\mathcal{R}}{4}\left(|A_s(t)|^2 + |A_{LO}|^2 + 2A_s(t)A_{LO}\cos\left(\overbrace{|\omega_s - \omega_{LO}|}^{\omega_{IF}}t + \theta_s(t)\right)\right)$

$I_2 = \frac{\mathcal{R}}{4}\left(|A_s(t)|^2 + |A_{LO}|^2 - 2A_s(t)A_{LO}\cos(\omega_{IF}t + \theta_s(t))\right)$

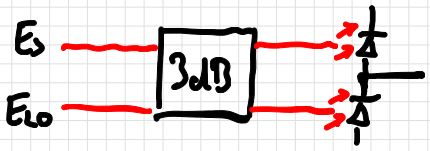
$I_3 = \frac{\mathcal{R}}{4}\left(|A_s(t)|^2 + |A_{LO}|^2 - 2A_s(t)A_{LO}\sin(\omega_{IF}t + \theta_s(t))\right)$

$I_4 = \frac{\mathcal{R}}{4}\left(|A_s(t)|^2 + |A_{LO}|^2 + 2A_s(t)A_{LO}\sin(\omega_{IF}t + \theta_s(t))\right)$

in-phase component: $I_I = I_1 - I_2 = \mathcal{R}A_s(t)A_{LO}\cos(\omega_{IF}t + \theta_s(t)) = \mathcal{R}A_s(t)A_{LO}\cos\theta_s(t)$ ($\omega_{IF} = 0$)

quadrature component: $I_Q = I_4 - I_3 = \mathcal{R}A_s(t)A_{LO}\sin\theta_s(t)$

g) 4-QAM <=> QPSK -> phase modulation, i.e. $A_s(t) = A_s = \text{const.}$



$$I = 2\pi A_s A_{LO} \cos \theta_s(t)$$

Problem 22

a) Bit rate: $R_b = 2 \cdot R_s \cdot \log_2 M \frac{1}{1+OH}$ ($R_s \hat{=} \Delta f$)

- 2: two polarizations
- $\log_2 M$: number of symbols
- $\frac{1}{1+OH}$: overhead from code

From diagram:

Code	pre-FEC BER	Modulation scheme
1	$2 \cdot 10^{-5}$	4-QAM
2	$1 \cdot 10^{-3}$	16-QAM
3	$3 \cdot 10^{-2}$	64-QAM

$$\Rightarrow R_{b,1} = 2 \cdot 50\,GBd \cdot \log_2 4 \frac{1}{1+0.03} = 194.2 \frac{Gbit}{s}$$

$$R_{b,2} = 2 \cdot 50\,GBd \cdot \log_2 16 \frac{1}{1+0.04} = 373.8 \frac{Gbit}{s}$$

$$R_{b,3} = 2 \cdot 50\,GBd \cdot \log_2 64 \frac{1}{1+0.2} = 500 \frac{Gbit}{s}$$

→ maximum bit rate at 64-QAM

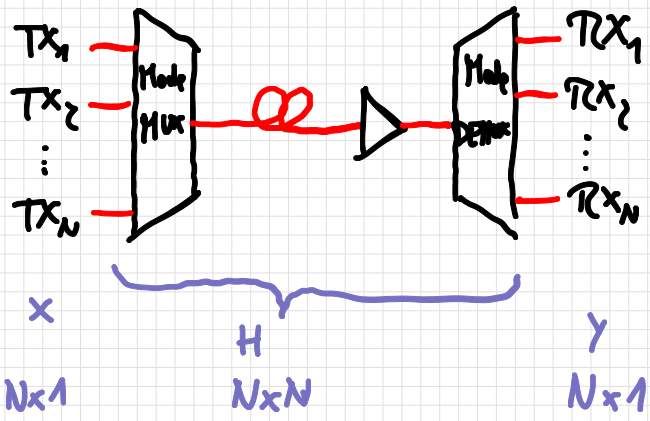
b) $C = \log_2(1+SNR)$ ... Shannon capacity

For $R_b < C \cdot \Delta f$, it is possible to reduce the BER with an arbitrary and suitable code in a transmission link. If $R_b \geq C \cdot \Delta f$, no such code exists.

$$C_{max} = \log_2(1+10^{12dB/10}) = 6 \frac{bit/s}{Hz}$$

$$\Rightarrow R_{b,max} = C_{max} \cdot \Delta f \cdot 2 = 2 \cdot 6 \frac{bit/s}{Hz} \cdot 50\,GHz = 600 \frac{Gbit}{s}$$

c) Transmission with few-mode fiber:



$$\Rightarrow y = Hx \quad \Leftrightarrow x = H^{-1} y$$

$$C = \sum_{i=1}^{N} \log_2(1 + \lambda_i \cdot SNR) \text{ with } \lambda_i \text{... eigenvalues } HH^T$$

N=3:

$$H \cdot H^T = \begin{pmatrix} 1 & 0.2 & 0.1 \\ 0.2 & 0.95 & 0.3 \\ 0.1 & 0.3 & 0.9 \end{pmatrix} \begin{pmatrix} 1 & 0.2 & 0.1 \\ 0.2 & 0.95 & 0.3 \\ 0.1 & 0.3 & 0.9 \end{pmatrix} = \begin{pmatrix} 1.05 & 0.42 & 0.25 \\ 0.42 & 1.0325 & 0.575 \\ 0.25 & 0.575 & 0.91 \end{pmatrix}$$

Eigenvalue calculations $\det \begin{pmatrix} 1.05-\lambda & 0.42 & 0.25 \\ 0.42 & 1.0325-\lambda & 0.575 \\ 0.25 & 0.575 & 0.91-\lambda \end{pmatrix} \overset{!}{=} 0$

$$\lambda^3 - 2.9925 \cdot \lambda^2 + 2.4094 \cdot \lambda - 0.5351 \overset{!}{=} 0$$

$$\Rightarrow \lambda_1 = 1.847, \lambda_2 = 0.7763, \lambda_3 = 0.3742$$

$$\Rightarrow C = \log_2(1 + 1.847 \cdot 10^{18dB/10}) + \log_2(1 + 0.7763 \cdot 10^{18dB/10}) + \log_2(1 + 0.3742 \cdot 10^{18dB/10})$$

$$= 14.14 \text{ bit/s/Hz}$$

Without crosstalk: $H = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow \lambda_i = 1 \Rightarrow C = 3 \cdot \log_2(1 + 10^{18dB/10}) = 18 \text{ bit/s/Hz}$

d) $y = Hx \Rightarrow x = H^{-1} y$

$\Rightarrow H^{-1}$ needs to be calculated

$$\left(H \;\middle|\; \begin{matrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{matrix} \right) \hookrightarrow \left(\begin{matrix} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{matrix} \;\middle|\; H^{-1} \right)$$ valid for arbitrary N

$N=2$:

$$H^{-1} = \frac{1}{\det H} \begin{pmatrix} h_{22} & -h_{12} \\ -h_{21} & h_{11} \end{pmatrix}$$

$N=3$:

$$H^{-1} = \frac{1}{\det H} \begin{pmatrix} h_{22}h_{33} - h_{23}h_{32} & h_{13}h_{32} - h_{12}h_{33} & h_{12}h_{23} - h_{13}h_{22} \\ h_{23}h_{31} - h_{21}h_{33} & h_{11}h_{33} - h_{13}h_{31} & h_{13}h_{21} - h_{11}h_{23} \\ h_{21}h_{32} - h_{22}h_{31} & h_{12}h_{31} - h_{11}h_{32} & h_{11}h_{22} - h_{12}h_{21} \end{pmatrix}$$

$$x = H^{-1} \cdot y = \begin{pmatrix} 1.0458 & -0.2051 & -0.0478 \\ -0.2051 & 1.2167 & -0.3828 \\ -0.0478 & -0.3828 & 1.244 \end{pmatrix} \begin{pmatrix} 0.3 \\ 1.75 \\ 1.2 \end{pmatrix} = \begin{pmatrix} 1 \cdot 10^{-6} \\ 0.99999 \\ 0.99996 \end{pmatrix} \approx \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$